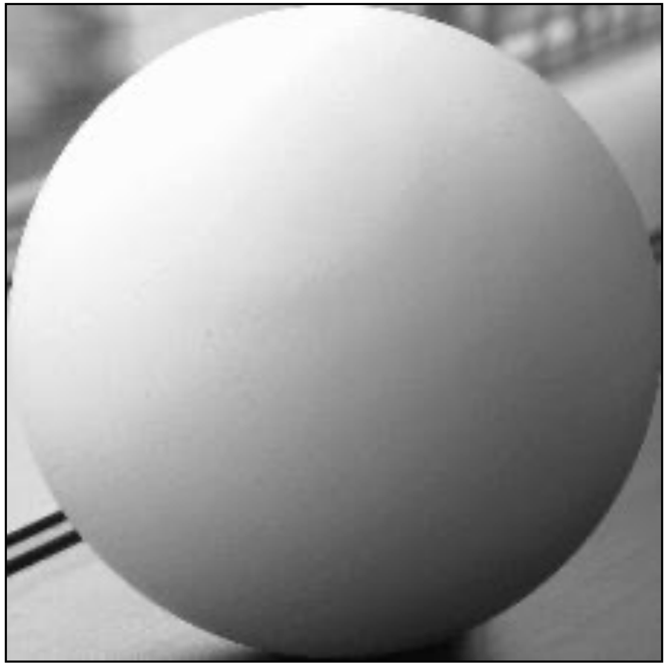
Shape from Shading

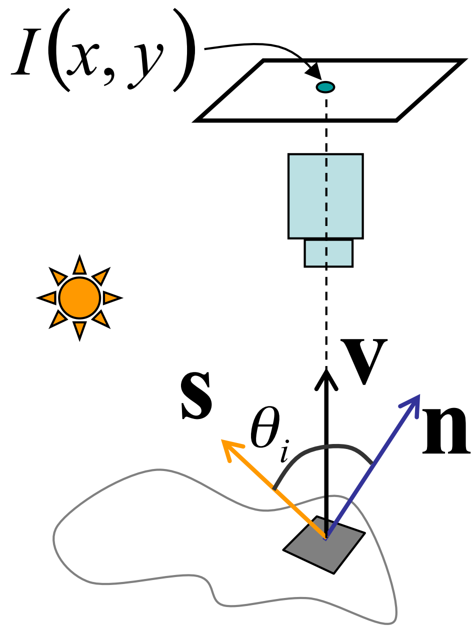
Image Intensity and 3D Geometry



- Shading as a cue for shape reconstruction
- What is the relation between intensity and shape?
  - Reflectance Map

Reflectance Map - RECAP

- Relates image irradiance *I(x,y)* to surface orientation *(p,q)* for given source direction and surface reflectance
- Lambertian case:



k : source brightness

ρ : surface albedo (reflectance)

c : constant (optical system)

Image irradiance:

$$I = \frac{\rho}{\pi} kc \cos\theta_i = \frac{\rho}{\pi} kc\, \mathbf{n} \cdot \mathbf{s}$$

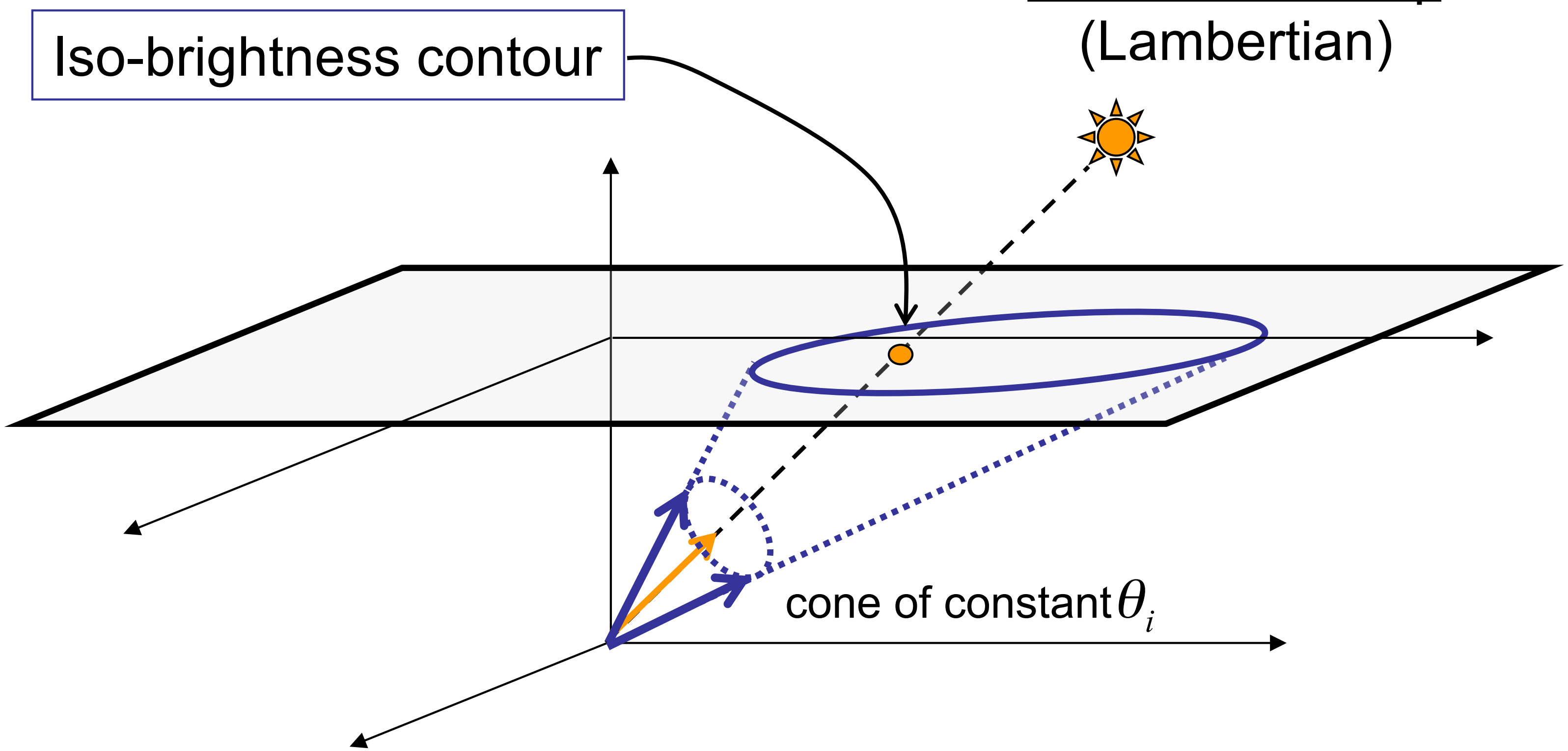
Let $\frac{\rho}{\pi} kc = 1$ then $\boxed{I = \cos\theta_i = \mathbf{n} \cdot \mathbf{s}}$

Reflectance Map - RECAP

- Lambertian case

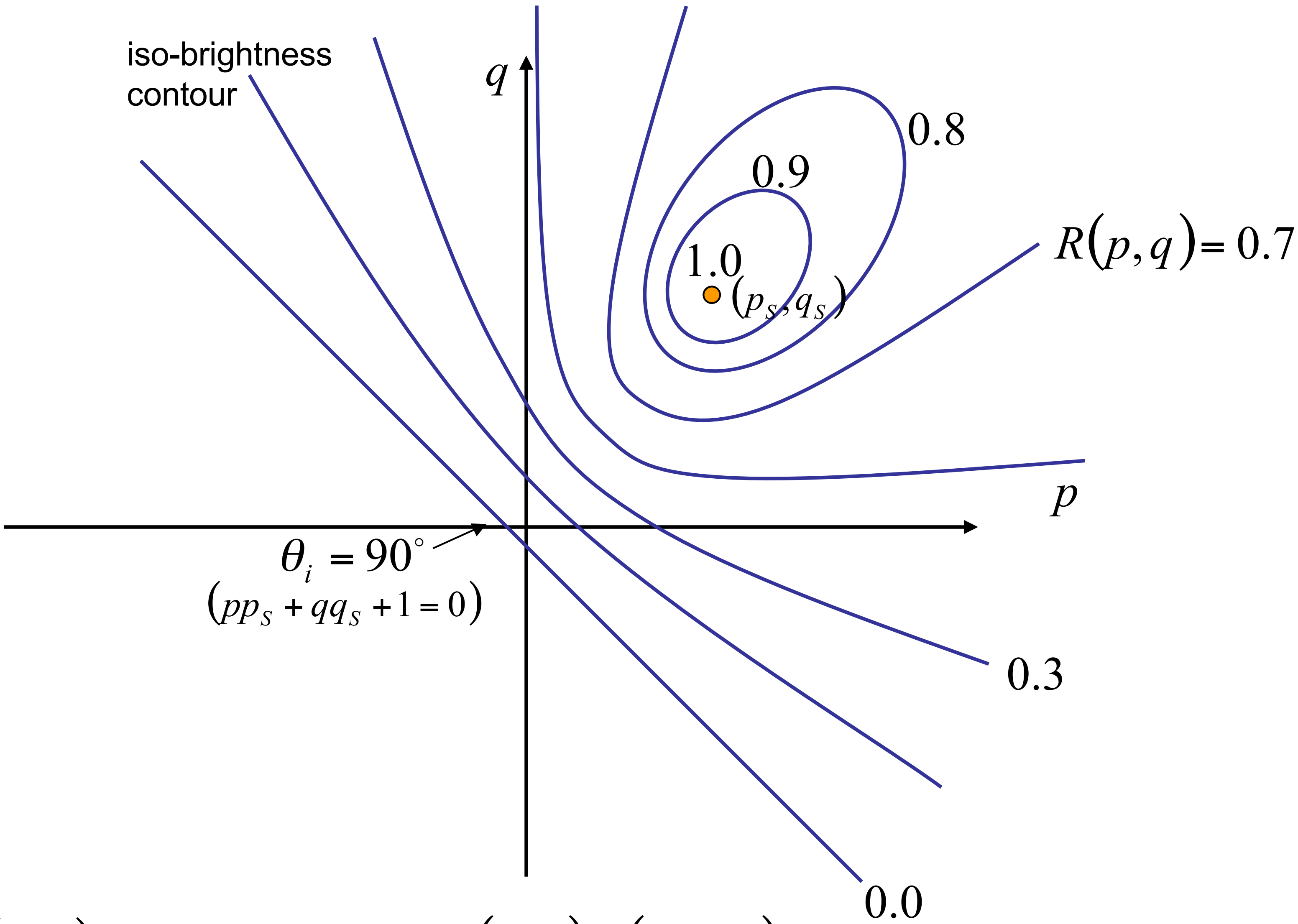
$$I = \cos\theta_i = \mathbf{n}\cdot\mathbf{s} = \frac{(pp_s + qq_s + 1)}{\sqrt{p^2+q^2+1}\sqrt{p_S^2+q_S^2+1}} = \boxed{R(p,q)}$$

Reflectance Map
(Lambertian)



Reflectance Map - RECAP

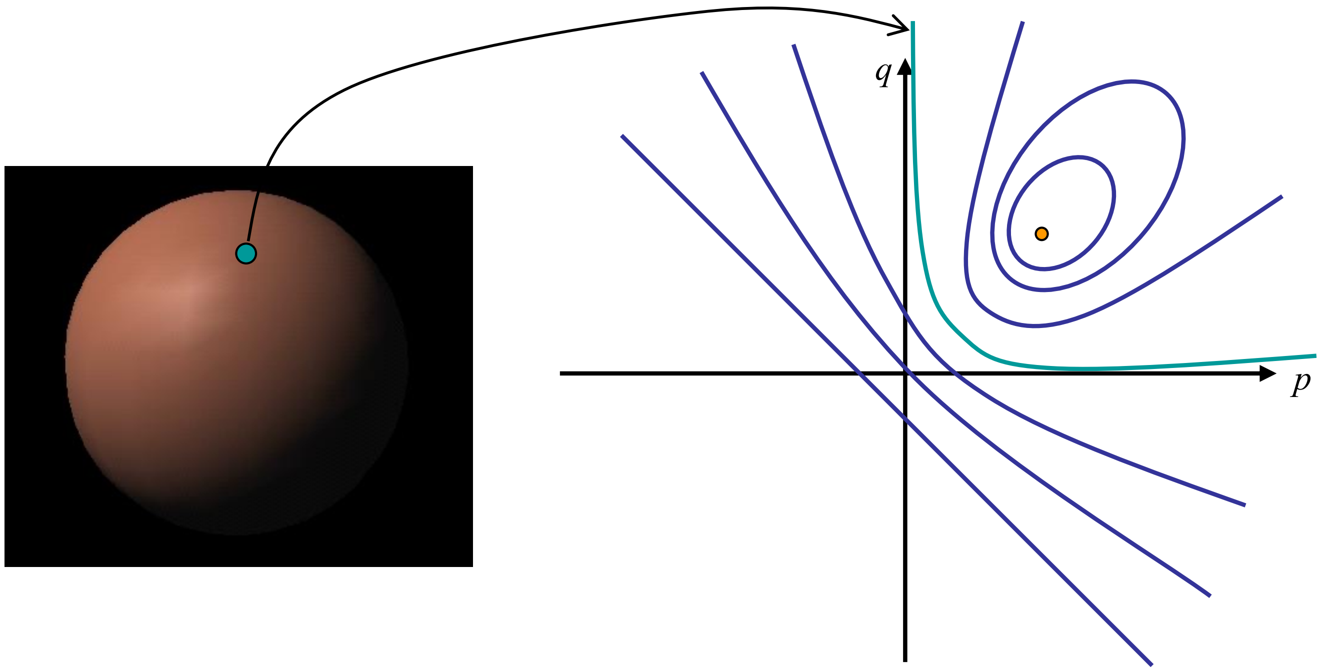
- Lambertian case



Note: $R(p,q)$ is maximum when $(p,q) = (p_S, q_S)$

Shape from a Single Image?

- Given a single image of an object with known surface reflectance taken under a known light source, can we recover the shape of the object?
- Given *R(p,q)* ((p_S,q_S) and surface reflectance) can we determine *(p,q)* uniquely for each image point?



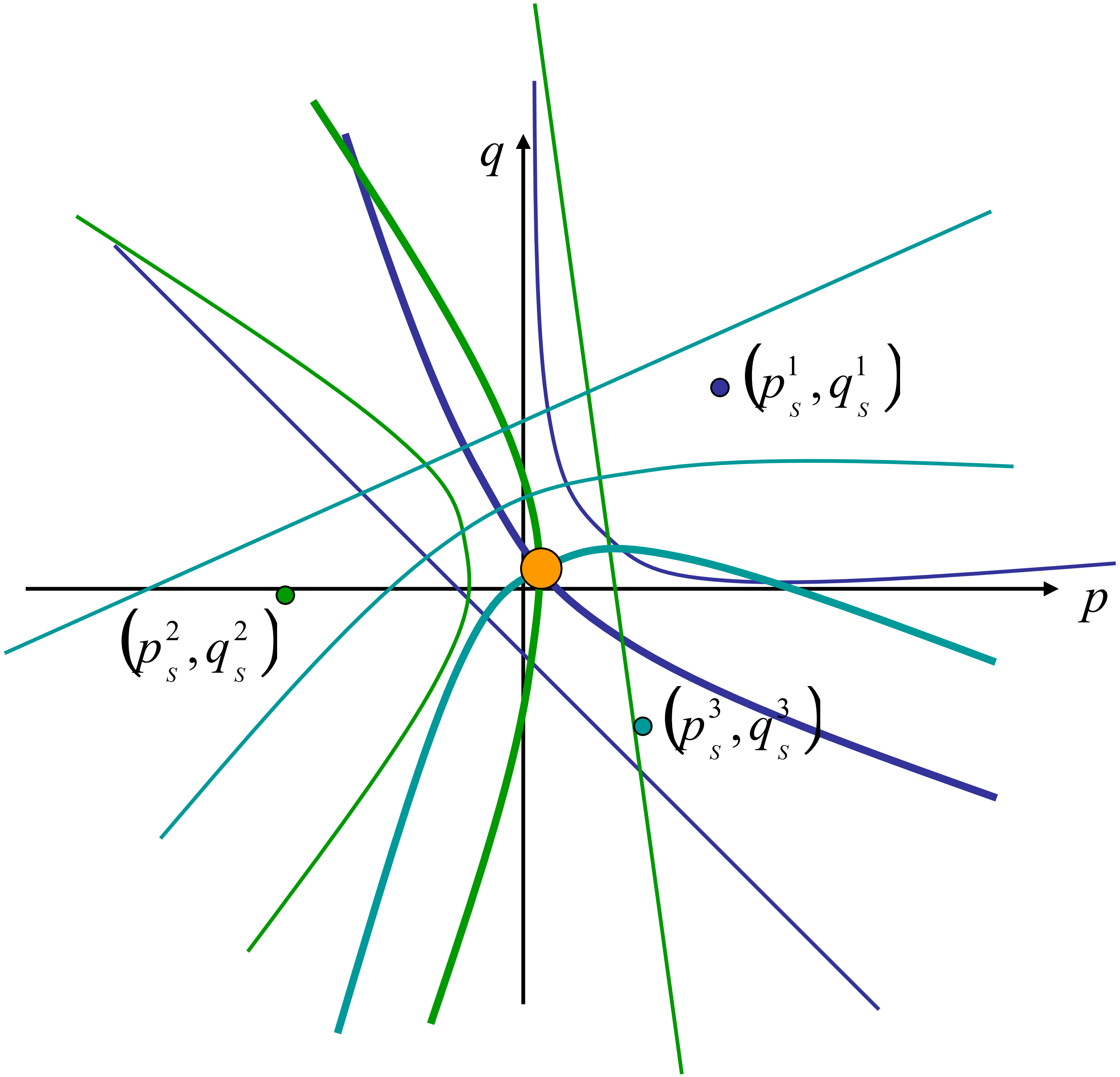
NO

Solution

- Take more images
  - Photometric stereo (previous class)

- Add more constraints
  - Shape-from-shading (this class)

Photometric Stereo

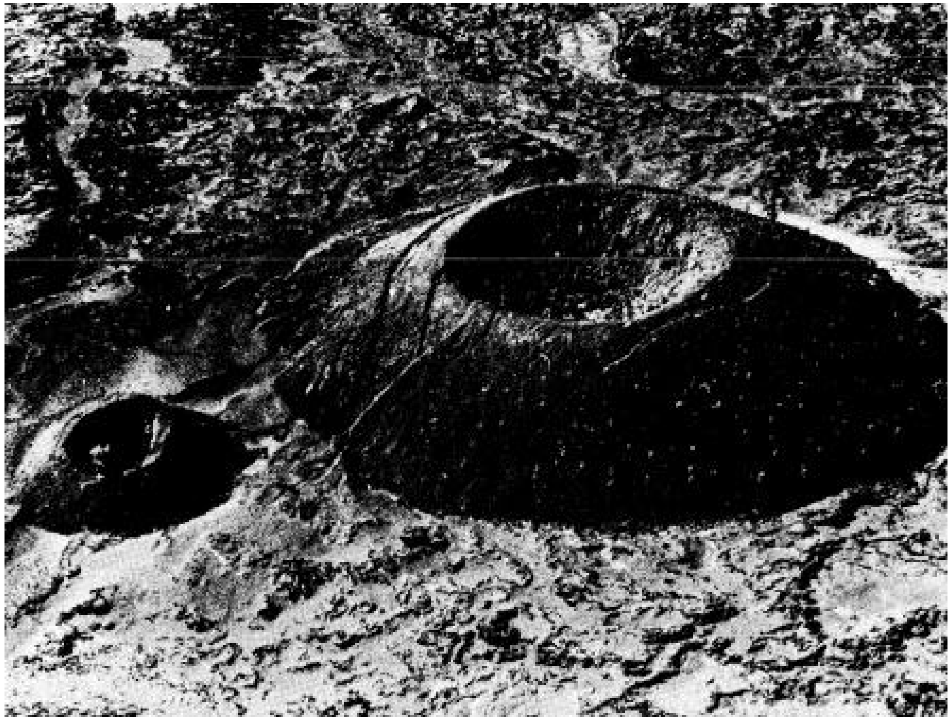


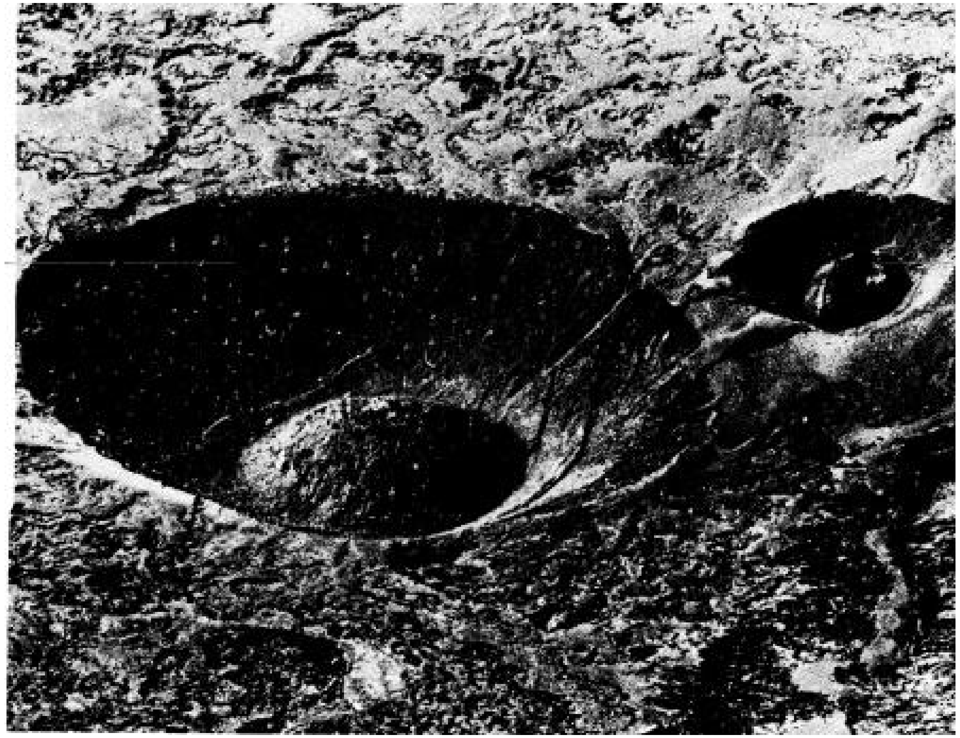
Solution

- Take more images
  - Photometric stereo (previous class)

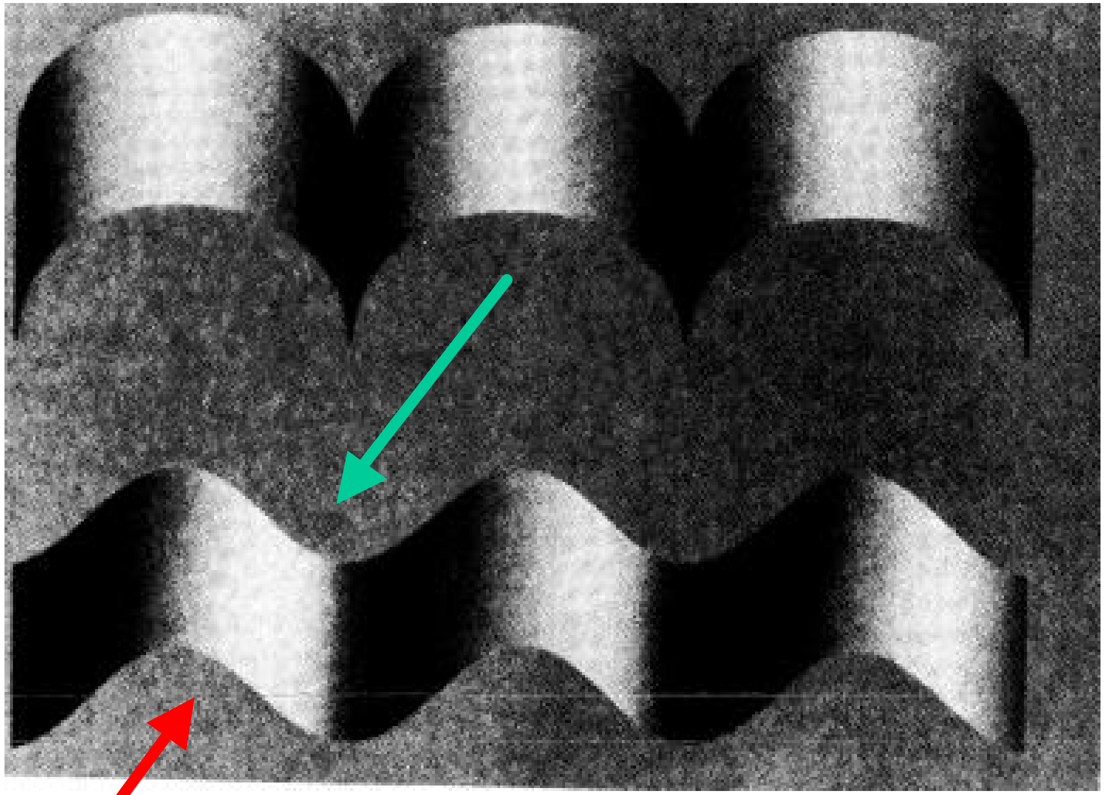
- Add more constraints
  - Shape-from-shading (this class)

Human Perception



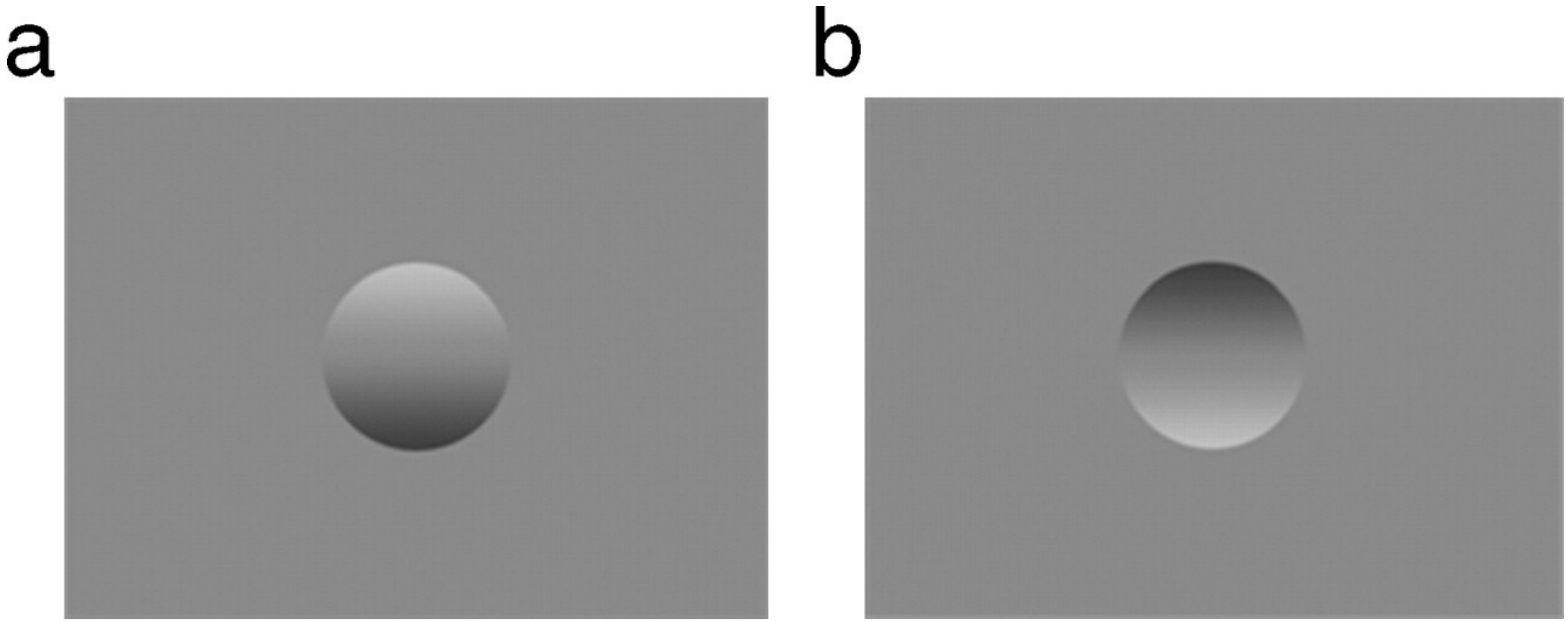


Human Perception



2 possible illumination hypotheses

Examples of the classic bump/dent stimuli used to test lighting assumptions when judging shape from shading, with shading orientations (a) 0° and (b) 180° from the vertical.



Thomas R et al. J Vis 2010;10:6

Human Perception

- Our brain often perceives shape from shading.
- Mostly, it makes many assumptions to do so.
- For example:

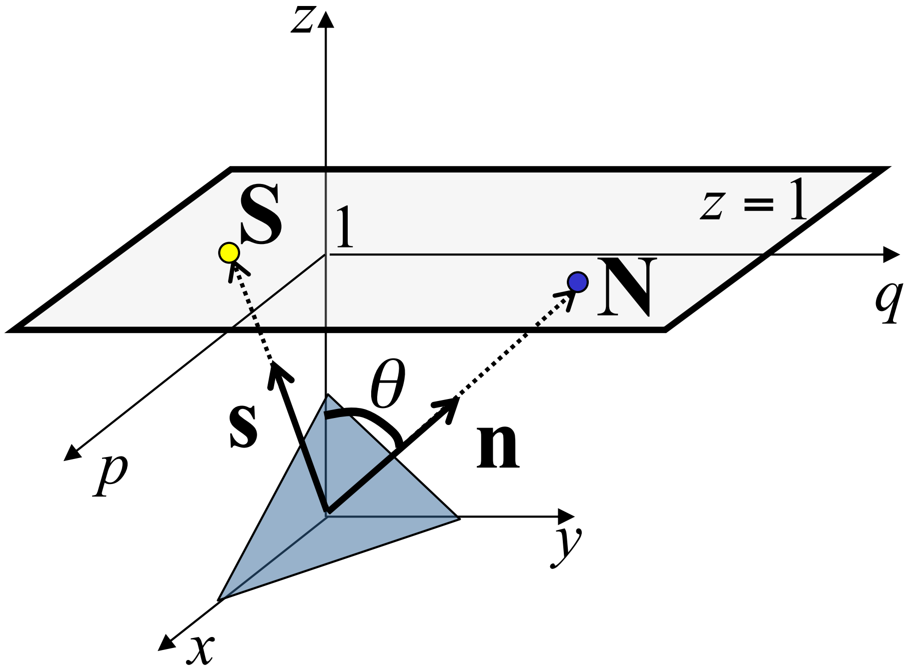
  Light is coming from above (sun).

  Biased by occluding contours.

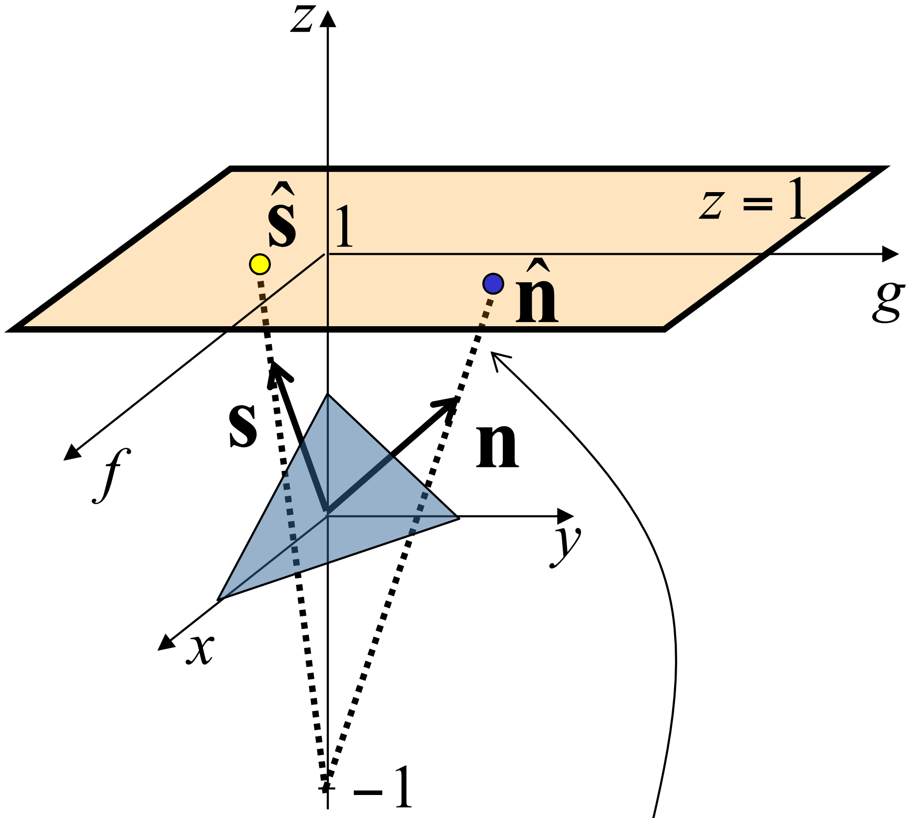
by V. Ramachandran

Stereographic Projection

(p,q)-space (gradient space)



(f,g)-space



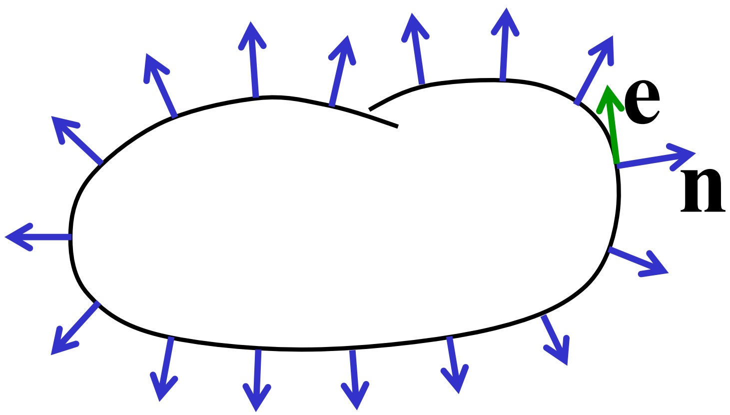
Problem

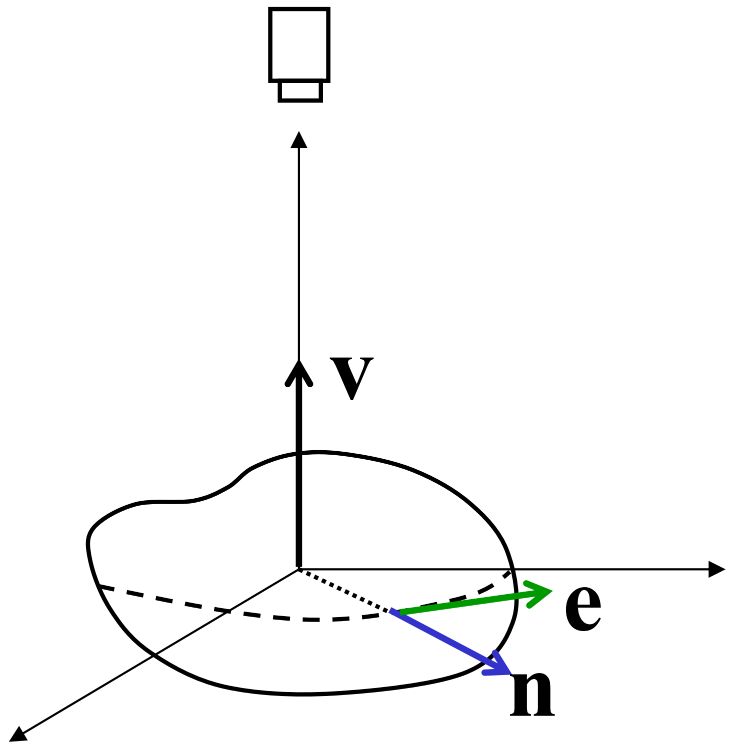
(p,q) can be infinite when $\theta = 90^\circ$

$$f = \frac{2p}{1+\sqrt{1+p^2+q^2}} \qquad g = \frac{2q}{1+\sqrt{1+p^2+q^2}}$$

Redefine reflectance map as $R(f, g)$

Occluding Boundaries





$\mathbf{n} \perp \mathbf{e}, \quad \mathbf{n} \perp \mathbf{v} \;\therefore\; \mathbf{n} = \mathbf{e} \times \mathbf{v}$ $\quad$ $\mathbf{e}$ and $\mathbf{v}$ are known

The $\mathbf{n}$ values on the occluding boundary can be used as the boundary condition for shape-from-shading

Image Irradiance Constraint

- Image irradiance should match the reflectance map

Minimize

$$\boxed{e_i = \iint_{\text{image}} \left(I(x,y) - R(f,g)\right)^2 dx\,dy}$$

(minimize errors in image irradiance in the image)

Smoothness Constraint

- Used to constrain shape-from-shading
- Relates orientations *(f,g)* of neighboring surface points

Minimize

$$e_s = \iint_{\text{image}} \left(f_x^2 + f_y^2\right) + \left(g_x^2 + g_y^2\right) dx\,dy$$

(f, g): surface orientation under stereographic projection

$$f_x = \frac{\partial f}{\partial x}, f_y = \frac{\partial f}{\partial y}, g_x = \frac{\partial g}{\partial x}, g_y = \frac{\partial g}{\partial y}$$

(penalize rapid changes in surface orientation *f* and *g* over the image)

Shape-from-Shading

- Find surface orientations *(f,g)* at all image points that minimize

$$e = \underbrace{e_s}_{\text{smoothness constraint}} + \overbrace{\lambda}^{\text{weight}} \underbrace{e_i}_{\text{image irradiance error}}$$

Minimize

$$\boxed{e = \iint_{\text{image}} \left(f_x^2 + f_y^2\right) + \left(g_x^2 + g_y^2\right) + \lambda\left(I(x,y) - R(f,g)\right)^2 dx\,dy}$$

Numerical Shape-from-Shading

- **Smoothness error** at image point *(i,j)*

$$s_{i,j} = \frac{1}{4}\left(\left(f_{i+1,j} - f_{i,j}\right)^2 + \left(f_{i,j+1} - f_{i,j}\right)^2 + \left(g_{i+1,j} - g_{i,j}\right)^2 + \left(g_{i,j+1} - g_{i,j}\right)^2\right)$$

Of course you can consider more neighbors (smoother results)

- **Image irradiance error** at image point *(i,j)*

$$r_{i,j} = \left(I_{i,j} - R\left(f_{i,j}, g_{i,j}\right)\right)^2$$

Find $\{f_{i,j}\}$ and $\{g_{i,j}\}$ that minimize

$$e = \sum_i \sum_j \left(s_{i,j} + \lambda r_{i,j}\right)$$

(Ikeuchi & Horn 89)

Numerical Shape-from-Shading

Find $\{f_{i,j}\}$ and $\{g_{i,j}\}$ that minimize $e = \sum_i \sum_j \left(s_{i,j} + \lambda r_{i,j}\right)$

If $f_{k,l}$ and $g_{k,l}$ minimize e, then $\dfrac{\partial e}{\partial f_{k,l}} = 0, \dfrac{\partial e}{\partial g_{k,l}} = 0$

$$\frac{\partial e}{\partial f_{k,l}} = 2\left(f_{k,l} - \bar{f}_{k,l}\right) - 2\lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\left.\frac{\partial R}{\partial f}\right|_{f_{k,l}} = 0$$

$$\frac{\partial e}{\partial g_{k,l}} = 2\left(g_{k,l} - \bar{g}_{k,l}\right) - 2\lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\left.\frac{\partial R}{\partial g}\right|_{g_{k,l}} = 0$$

where $\bar{f}_{k,l}$ and $\bar{g}_{k,l}$ are 4-neighbors average around image point *(k,l)*

$$\bar{f}_{k,l} = \frac{1}{8}\left(f_{i+1,j} + f_{i,j+1} + f_{i-1,j} + f_{i,j-1}\right)$$

$$\bar{g}_{k,l} = \frac{1}{8}\left(g_{i+1,j} + g_{i,j+1} + g_{i-1,j} + g_{i,j-1}\right)$$

(Ikeuchi & Horn 89)

Numerical Shape-from-Shading

$$\frac{\partial e}{\partial f_{k,l}} = 2\left(f_{k,l} - \bar{f}_{k,l}\right) - 2\lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\frac{\partial R}{\partial f}\bigg|_{f_{k,l}} = 0 \qquad \frac{\partial e}{\partial g_{k,l}} = 2\left(g_{k,l} - \bar{g}_{k,l}\right) - 2\lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\frac{\partial R}{\partial g}\bigg|_{g_{k,l}} = 0$$

Update rule

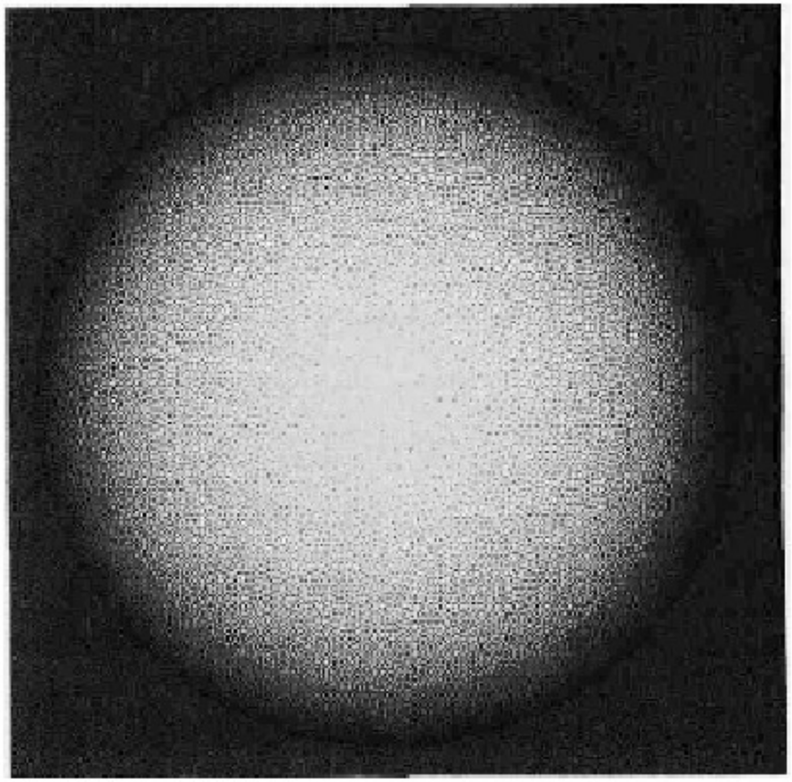
$$f_{k,l}^{n+1} = \bar{f}_{k,l}^{n} + \lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\frac{\partial R}{\partial f}\bigg|_{f_{k,l}}$$

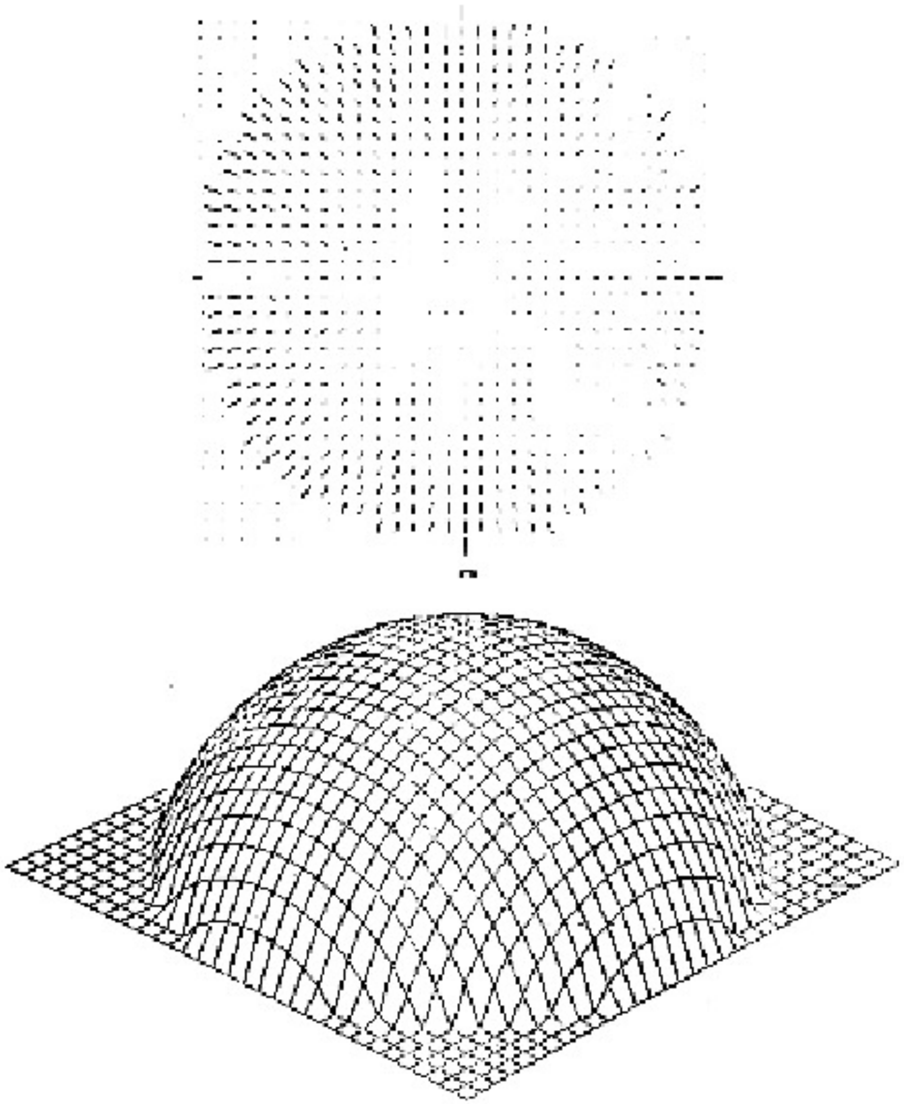
$$g_{k,l}^{n+1} = \bar{g}_{k,l}^{n} + \lambda\left(I_{k,l} - R\left(f_{k,l}, g_{k,l}\right)\right)\frac{\partial R}{\partial g}\bigg|_{g_{k,l}}$$

- Use known (f, g) values on the occluding boundary to constrain the solution (boundary conditions)
- Compare $\left(f_{k,l}^{n+1}, g_{k,l}^{n+1}\right)$ with $\left(f_{k,l}^{n}, g_{k,l}^{n}\right)$ for convergence test
- As the solution converges, increase λ to remove the smoothness constraint

(Ikeuchi & Horn 89)

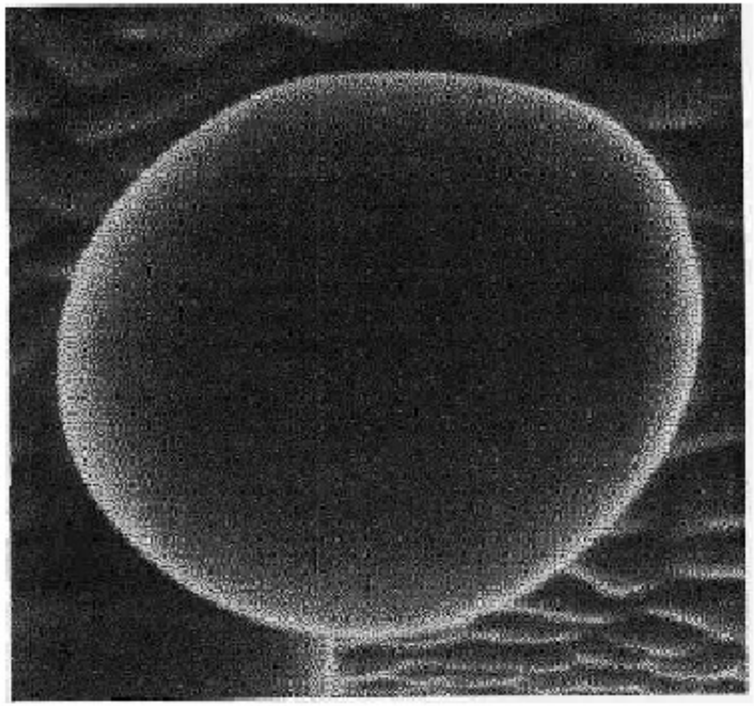
Results



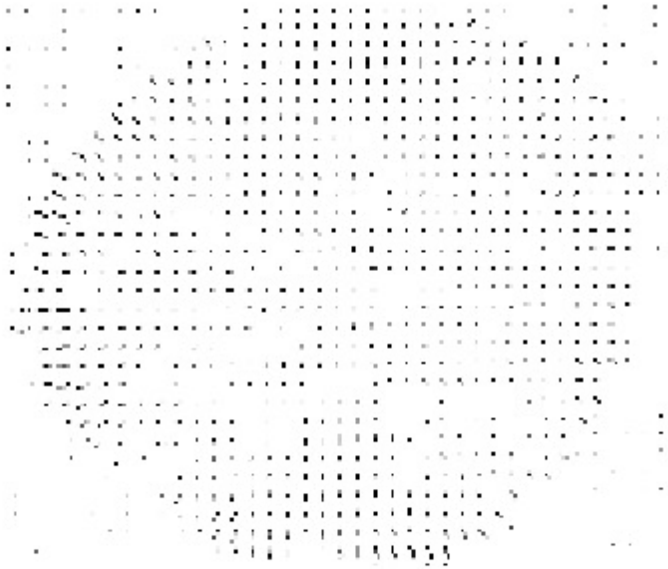


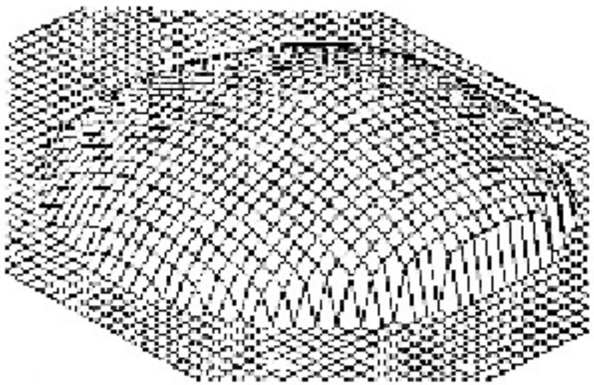
by Ikeuchi and Horn

Results



Scanning Electron Microscope image
(inverse intensity)





by Ikeuchi and Horn

Next Part

- Geometry
  - Image projection
  - Motion and Tracking
  - Stereo
  - Range image sensors